INCH-POUND

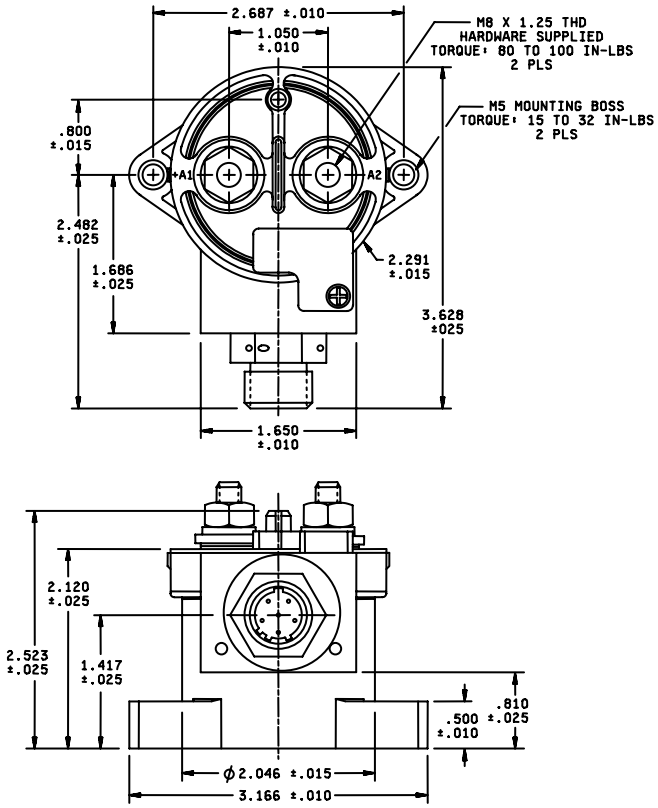
MIL-PRF-32085/3
11 May 2006

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, SPST, 270 VOLT DC,
300 AMPERE HERMETICALLY SEALED,
TRANSIENT SUPPRESSED DC COIL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-32085.



mm	Inches	mm
0.51	.800	20.32
1.52	1.05	26.7
2.5	1.125	28.58
3.8	1.380	35.05
3.81	1.810	45.97
5.46	1.927	48.95
5.61	2.25	57.2
6.35	2.36	59.9
9.4	3.37	85.6
10.2	3.48	88.4
12.75	3.88	98.6
14.38	12.0	305
15.62		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on all dimensions is ±.030 inch (0.77 mm).

FIGURE 1. Dimensions and configurations.

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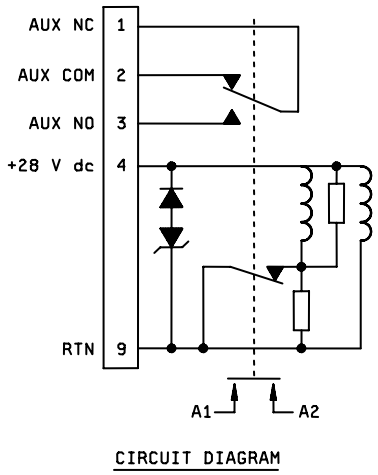


FIGURE 2. Circuit diagram.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Terminations: See figure 1.

Configuration: 1PST: Electrically held.

Weight: 0.8 pound maximum.

Seal: Hermetic

Finish: Flat black or off-white.

Rated coil voltage: 28 V dc.

Maximum coil voltage: 32 V dc.

Pickup voltage:

Nominal: 16 V dc at 25°C.

Nominal: 18 V dc at -40°C to +71°C.

High temperature test: 19 V dc.

Continuous current test: 22 V dc.

Dropout voltage: 1.5 V dc minimum.

Hold voltage: 10 V dc minimum, -40°C to +71°C.

Voltage rating:

Main contacts: 270 V dc

Auxiliary contacts 28 V dc.

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Current rating:

Main contacts:

Resistive: 300 amperes.

Inductive: 150 amperes.

Motor: 75 amperes.

Auxiliary contacts:

Resistive: 0.1 amperes at 28 V dc.

Coil resistance at 25°C, ±10 percent.

Pickup: 5.7 ohms.

Pickup + hold: 104 ohms.

Coil current:

Pickup: 3.8A peak maximum, -40°C to +71°C.

Holding: 0.5A maximum, -40°C to +71°C.

Contact voltage drop:

Before life: 100 mV maximum.

End of life: 150 mV maximum.

Operate time: 40 ms maximum.

Release time: 15 ms maximum.

Contact bounce time (main and auxiliary): 5 ms maximum.

Duty rating:

Main and auxiliary contacts: Continuous.

Coil: Continuous.

Overload current: 500 amperes dc.

Rupture current: 1,500 amperes dc, 25 cycles.

Coil transient suppression (electrically held): 42 V dc maximum.

Life: Resistive, motor, and inductive, 25,000 cycles each.

Mechanical life: 100,000 cycles.

Dielectric withstanding voltage:

Sea level: 1,500 V rms from main contacts (closed) to all other points.

70,000 feet: 500 V rms from main contacts (closed) to all other points.

Temperature range:

Operating: -40°C to +71°C.

Non-operating: -55°C to +85°C.

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Vibration:

Sinusoidal: 15g in accordance with method 204 of MIL-STD-202.

Frequency range: 10 to 2,000 Hz.

Random: 8.5g rms overall, in accordance with method 214 of MIL-STD-202.

Frequency (Hz)	PSD(g^2/Hz)
10	.01
10-50	+3 dB/Oct
50-1,000	.05
1,000-2,000	-7 dB/Oct

Acceleration: 15g.

Maximum altitude rating (operating): 70,000 feet.

Shock: 20g's in accordance with method 213 of MIL-STD-202.

Marking: Marking shall be in accordance with MIL-PRF-32085.

Part or Identifying Number (PIN): M32085/3- (dash number from table I and suffix letter designating failure rate level.)

TABLE I. Dash numbers and type of loads tested during life.

Dash number	Type of loads tested during life
100	Resistive
101	Resistive and motor
110	Resistive and inductive
111	Resistive, inductive, and motor

Referenced documents. In addition to MIL-PRF-32085 this document references the following:

MIL-STD-202

Custodians:
Army - CR
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2006-011)

Review activities:
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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